

INTERNSHIP FINAL REPORT

Student Name: Kratika Gupta

Company: ShadowFox

University: College of Technology and Engineering, Udaipur (CTAE)

Role: Data Analyst

Internship Duration: August 1st, 2026 – August 31st, 2026

Mentor: Mr. Aakash

Project Dataset & Dashboard: https://docs.google.com/spreadsheets/d/19U3cgfW8_Mp346jUasA2Hw4F-C2-rLzmqGXA9SpikLQ/edit?usp=drivesdk

Objectives

The primary objective of this Data Analyst internship track at **ShadowFox** was to evaluate and enhance core competency in practical data understanding, spreadsheet-based analytics, visual dashboard design, and business-oriented decision reporting.

The specific objectives achieved during the internship program include:

- **Data Preprocessing & Hygiene:** Acquire, structure, clean, and validate raw operational/sales datasets within Google Sheets and Microsoft Excel.
- **Key Metric Formulation:** Identify and compute essential Key Performance Indicators (KPIs) such as Total Revenue, Gross Profit Margins, Average Order Value (AOV), Order Volumes, and Month-over-Month (MoM) Growth Rates.
- **Multi-Dimensional Trend Analysis:** Perform granular analysis across multiple dimensions including temporal movements (monthly trends), product category distributions, top-performing vs. low-performing SKUs, and regional sales variations.
- **Interactive Dashboard Construction:** Design and assemble a clean, visual spreadsheet dashboard incorporating summary scorecards, trend visualizers, pivot tables, and dynamic filtering controls (slicers).
- **Actionable Business Insights:** Translate complex numerical data into clear, non-technical executive summaries and strategic recommendations aimed at optimizing business performance.

Tasks and Responsibilities

During the 1-month Data Analyst internship, key responsibilities were executed progressively across structured analytical phases:

1. Data Cleaning & Preparation

- **Import & Structuring:** Loaded raw business sales and transaction data into Google Sheets and conducted initial structural audits.
- **Handling Data Quality Issues:** Identified and removed duplicate records, standardized text strings using TRIM() and PROPER(), and addressed blank entries using logical handling routines like IFERROR().
- **Data Type Standardization:** Converted raw text dates into unified ISO formats (YYYY-MM-DD), formatted currency variables, and verified transaction key uniqueness.

2. Quantitative Analysis & Metric Selection

- **Formulas & Functions:** Implemented robust lookup and conditional logic using SUMIFS, COUNTIFS, AVERAGEIFS, XLOOKUP, INDEX-MATCH, and date parsing functions.
- **KPI Scorecard Generation:** Formulated high-level business health metrics: Total Sales Revenue, Gross Profit Margin, Total Orders, Average Order Value (AOV), and Month-over-Month (MoM) Growth (%).

3. Dimensional Analysis & Segmentation

- **Monthly Movement Analysis:** Aggregated monthly revenue and transaction volume to detect sales seasonality, holiday demand surges, and off-peak declines.
- **Category & Product Breakdown:** Segmented sales by product line to isolate top-revenue generators (hero items) from lagging, low-margin product categories.
- **Regional Performance Analysis:** Compared performance across geographic territories to highlight regional growth drivers and logistical bottlenecks.

4. Visual Dashboard & UI Design

- Constructed an executive **Google Sheets Dashboard** featuring KPI summary cards, line charts for monthly trends, bar/column charts for category breakdowns, donut charts for regional market share, and interactive slicers for dynamic filtering.

Learning Outcomes

- **Advanced Spreadsheet Analytics:** Mastered Google Sheets & Excel analytical capabilities (nested formulas, dynamic pivot tables, array handling, and conditional formatting rules).
- **End-to-End Data Pipeline Management:** Developed a methodical approach to transform messy, unorganized raw data into clean, business-ready models.
- **Data Visualization & UX Layout:** Mastered best practices in dashboard ergonomics, color psychology, metric hierarchy, and visual storytelling.
- **Commercial & Business Literacy:** Gained a deep understanding of core commercial metrics and how operational performance maps to financial outcomes.
- **Executive Reporting & Communication:** Enhanced ability to present concise, actionable data insights tailored for managerial decision-making.

Challenges and Solutions

Challenge Encountered	Root Cause	Solution Implemented
Messy & Inconsistent Raw Data	Mixed date formats, trailing whitespaces, and missing	Applied automated data cleaning pipelines using TRIM(), PROPER(), date parsing, and

Challenge Encountered	Root Cause	Solution Implemented
	entries in transaction logs.	built-in Google Sheets data scrubbing utilities.
Visual Dashboard Overcrowding	Attempting to display too many metrics simultaneously created visual clutter.	Structured the dashboard into logical zones: KPI scorecards at top, macro trend graphs in center, and detailed pivot views at bottom with slicers.
Isolating Underperforming Drivers	Aggregate revenue obscured localized drop-offs in specific product categories.	Built dynamic multi-factor pivot tables and cross-filtering slicers to drill down into product-by-region combinations.

Conclusion

The **Data Analyst Internship** at **ShadowFox** provided an invaluable opportunity to bridge theoretical concepts with real-world business data analysis. By successfully performing data cleaning, KPI derivation, trend segmentation, and interactive dashboard creation in Google Sheets, all beginner-level track milestones were accomplished.

The final dashboard and analytical report demonstrate how raw operational data can be successfully converted into strategic, decision-oriented insights, providing a strong foundation for advanced analytics roles.

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ShadowFox Internship Program

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